The Basics of Capital Budgeting

- Capital budgeting is used by companies to evaluate major projects and investments, such as new plants or equipment.
- The process involves analyzing a project's cash inflows and outflows to determine whether the expected return meets a set benchmark.
- The major methods of capital budgeting include discounted cash flow, payback, and throughput analyses.

Generating Ideas for Capital Projects

The process starts by generating potential ideas for capital budgeting projects. These may be projects to improve existing processes within the firm (such as updating current manufacturing equipment or introducing new software to streamline our distribution) or they could be developing new product lines.

The most common examples of capital projects are infrastructure projects such as railways, roads, and dams. In addition, these projects include assets such as subways, pipelines, refineries, power plants, land, and buildings. Capital projects are also common in corporations.

Project Classifications

There are two types of Capital budgeting projects.

Mutually Exclusive Projects

Mutually exclusive projects are any set of projects in which choosing one makes the other projects no longer possible. For example, we are considering upgrading our printing press and have the choice of two alternatives. The first is a low-cost model that will need to be replaced in 3 years and the second is a more expensive model that will need to be replaced in 5 years. We can only choose one of these options, so they are mutually exclusive. When we have mutually exclusive projects, our decision rule needs to not only decide if a project is good or bad, but needs to be able to rank which project is the best.

Independent Projects

Independent (sometimes called stand-alone) projects are any set of projects in which choosing one has no impact on our decision to choose another project from that set. For example, Vantage Burgers Incorporation in Toronto, Canada Operated by M. Kashif Ali may have the following capital budgeting projects to consider. The first is a new deep frying system for their French fries. The second is a new order placement system for the drive-thru. Vantage Burgers Incorporation could choose to take the new deep fryer or the new order placement, or it could choose both. Taking one project does not influence the other, so they are independent. When we have independent projects, our decision rule does not need to rank which project is the best, but merely identify if the project is good or bad.

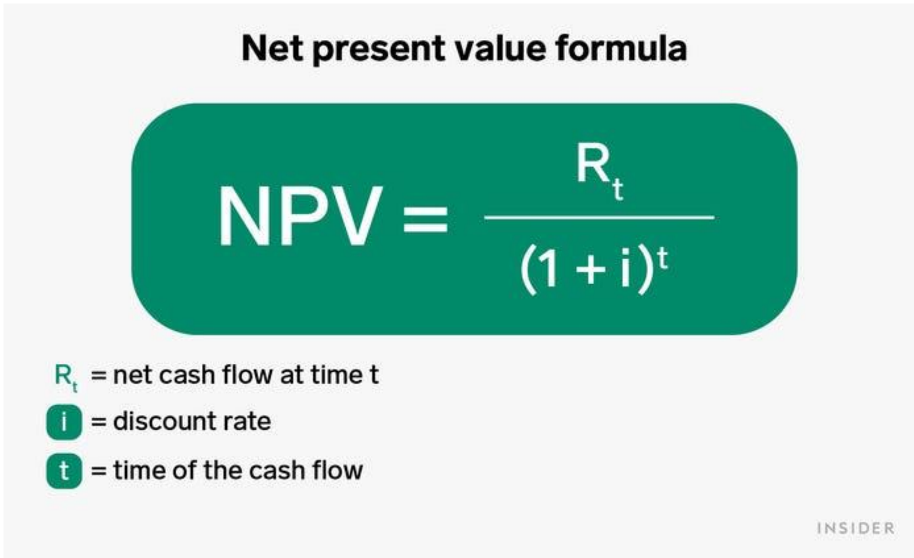
Types of Capital Investment Projects

1. New Products or New Markets.
2. Expansion of Existing Products or Markets.
3. Replacement Project Necessary to Continue Normal Operations.
4. Replacement Project Necessary to Reduce Business Costs

Net Present Value (NPV)

Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. NPV is used in capital budgeting and investment planning to analyze the profitability of a projected investment or project. NPV is the result of calculations used to find today's value of a future stream of payments. If the NPV of a project or investment is positive, it means that the discounted present value of all future cash flows related to that project or investment will be positive, and therefore attractive.

Formula of NPV



Let us say Nice Ltd wants to expand its business and so it is willing to invest Rs 10, 00,000. The investment is said to bring an inflow of Rs. 1,00,000 in the first year, 2,50,000 in the second year, 3,50,000 in the third year, 2,65,000 in the fourth year, and 4,15,000 in the fifth year. Assuming the discount rate to be 9%. Let us calculate NPV using the formula.

Year	Cash flow	Present Value
0	-10,00,000	-10,00,000
1	1,00,000	91,743
2	2,50,000	2,10,419
3	3,50,000	2,70,264

4	265,000	1,87,732
5	4,15,000	2,69,721

The cash inflow of Rs. 1, 00,000 at the end of the first year is discounted at the rate of 9% and the present value is calculated as Rs. 91,743. The cash inflow of Rs, 2, 50,000 at the end of year 2 is discounted and the present value is calculated as Rs. 2, 10,429, and so on. The total sum of the present value of cash inflows for all 5 years is Rs. 10, 29,879. The initial investment is Rs. 10, 00,000. Hence, the NPV is Rs. 29879. Since the NPV is positive the investment is profitable and hence Nice Ltd can go ahead with the expansion.

Internal Rate of Return (IRR)

- The internal rate of return (IRR) is the annual rate of growth that an investment is expected to generate.
- IRR is calculated using the same concept as net present value (NPV), except it sets the NPV equal to zero.
- IRR is ideal for analyzing capital budgeting projects to understand and compare potential rates of annual return over time.